



CALICUT UNIVERSITY – FOUR-YEAR UNDER GRADUATE PROGRAMME (CU-FYUGP)

BSc PHYSICS HONOURS

Programme	B.Sc. Physics Honours				
Course Title	INTRODUCTORY BIOPHYSICS				
Type of Course	Major Elective (SPECIALIZATION III: PHYSICS IN BIOLOGY)				
Semester	VI				
Academic Level	300 - 399				
Course Details	Credit	Lecture per week	Tutorial per week	Practical per week	Total Hours
	4	4	-	-	60
Pre-requisites	1. Higher Secondary level Physics 2. Fundamental Mathematics Concepts: Concept of calculus- Solution of very simplest differential equation 3. High school level Chemistry and Biology				
Course Summary	In this course the student learn a bridge between Physics and Biology. Look at some of the biological phenomena and analyze them with math and physics to gain important insights. This course tries to show that there is a quantitative, Physical sciences approach to Biological problems.				

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	Explain why is nano world so different from macro world and predict what's going on there by incorporating physical ideas like Random walk, Diffusion, probabilistic facts, etc.	U	C	Instructor-created exams, Assignments
CO2	Explain the biological systems and models by dealing with statistical mechanics and transport phenomena	U	C	Instructor-created exams

CO3	Answer many real life questions like why don't bacteria swim like fish by applying equation of motion appropriate to the nano world	Ap, U, R	P	Instructor-created exams, Assignments
CO4	Explain the thermodynamic basis of various biochemical reactions in cells and tissues.	R, U	F	Instructor-created exams
CO5	Analyse the role of action potential in nerve impulses, and the physics of signal communication via neural systems.	Ap, U	P	Instructor-created exams
CO6	Explain everyday phenomena and various processes in living systems by applying physical principles	An	C	Assignments/Seminar presentations
CO7	Make quantitative predictions by making a simplified model by applying many tools given in the course	An, C, U	M	Assignment/Group Projects/Presentations
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus:

Module	Unit	Content	Hrs (12+ 48)	Marks (70)
I	Open Ended Module		12	
	Lectures/Reading: (4 hrs) Introduction - Nature and Subject of Biophysics Chapter 1 from Book 2 Mysteries, Metaphors, Models- What the ancients knew? What's inside cells Heat-Heat is a form of energy, How life generates order, How to do better on exams (and discover new physical laws), Dimensional analysis can help you catch errors and recall definitions, Dimensional analysis can also help you formulate hypotheses What's inside cells Cell physiology Sections 1.1.1, 1.2.1, 1.4.1, 1.4.2, 1.4.3, 2.1 of Book 1			

	Simulation (Along with the following modules or after completion of modules) (4 hrs) Example: Generation of random numbers, say between 0 and 1, and draw the probability distribution and fit curve with Gaussian distribution. (Note 1: Use any programming language or software Note 2: Use any software to plot, like gnu plot.)			
	Open-Ended Exploration and Assessment: (4 hrs) Recent development in Biophysics: Read an article from a scientific journal and discuss in groups and present as seminar			
	Group Assignment: Design a biological system which reflects the application of theories from any of the modules II-V or Find out an open problem in Biological system where physics theories make role.			
II	The Molecular Dance		18	26
	1	Probabilistic facts of life- <i>Discrete distribution, Continuous distribution, Expectation and variance, addition and multiplication rules</i>	2	
	2	Decoding the ideal gas law- <i>Temperature reflects the average kinetic energy of thermal motion, The complete distribution of molecular velocities is experimentally measurable, Boltzmann distribution, Activation barriers and control reaction rates, Relaxation to equilibrium</i>	3	
	3	Statistics of genetics & heredity: historical example	2	
	4	Brownian motion- <i>Just a little more history, Random walks lead to diffusive behaviour</i>	2	
	5	<i>Diffusion law is model independent, Friction is quantitatively leads to diffusion- Einstein's relation</i>	2	
	6	Other random walks- <i>The Confirmation of polymers</i>	1	
	7	Diffusion rules the sub cellular world, Diffusion follows a differential equation- <i>Fick's law, Diffusion equation</i>	2	
	8	Precise statistical prediction of random processes, Functions, Derivatives, and snakes under the rug	2	
	9	Biological Applications of Diffusion- <i>The permeability of artificial membranes is diffusive, Diffusion sets a fundamental limits on bacterial metabolism</i>	2	
Sections 3.1, 3.2, 3.3, 4.1, 4.3.1, 4.4, 4.5, 4.6.1, 4.6.2 of Book 1				
III	Life in the slow lane: The low Reynolds number world- Why don't bacteria swim like fish?		10	14

	10	Friction in Fluids- <i>Sufficiently small particles can remain in suspension indefinitely, The rate of sedimentation depends on solvent viscosity, Its hard to mix a viscous liquid</i>	3	
	11	Low Reynolds number- <i>Viscous force in Newtonian fluid, Relative importance of friction and inertia, time-reversal properties of dynamical law and dissipative character</i>	3	
	12	Biological Applications- <i>Swimming and Pumping, To Stir or Not to Stir?, Foraging, Attack, and Escape</i>	2	
	13	<i>Vascular networks, Viscous drag at DNA replication fork</i>	2	
Sections 5.1, 5.2, 5.3, of Book 1				
IV	Entropy, Temperature, and Free energy		11	16
	14	How to measure disorder; Entropy- <i>The Statistical Postulate, Entropy is a constant times the maximal value of disorder</i>	2	
	15	Temperature- <i>Heat flows to maximise disorder, Temperature is a statistical property of a system in equilibrium</i>	2	
	16	The Second Law- <i>Entropy increases spontaneously when a constraint is removed, Three remarks</i>	2	
	17	Open Systems- Free energy of a subsystem reflects the competition between entropy and energy, Entropic forces as derivatives of free energy	3	
	18	Microscopic systems- The Boltzmann distribution follows from the statistical postulate, The minimum free energy principle also applies to microscopic systems	2	
Sections 6.1, 6.2, 6.3, 6.4, 6.5.1, 6.5.2, 6.6.1, 6.6.3 of Book 1				
V	Nerve Impulses		9	14
	19	The problem of nerve impulses- <i>Phenomenology of action potential</i>	2	
	20	Cell membrane as an electrical network	3	
	21	Simplified mechanism of action potential- <i>The puzzle, mechanical analogy</i>	2	
	22	Nerve, Muscle, synapse	2	
Sections 12.1.1, 12.1.2, 12.2.1, 12.2.2, 12.4 of Book 1				
Book for Study: 1. Biological Physics: Energy, Information, Life (<i>Student Edn.</i>) by Philip Nelson (Book 1) 2. Biophysics- An Introduction (<i>2nd Edn.</i>), Roland Glaser 3. Biophysics: An Introduction, 2nd Edn by Rodney Cotterill 4. Physical Biology of the Cell, R. Phillips, J. Kondev and J. Theriot				

5. Random Walks in Biology, Howard Berg
6. Zoological Physics: Quantitative Models of Body Design, Actions, and Physical Limitations of Animals by Boye K. Ahlborn

Mapping of COs with PSOs and POs :

	PSO 1	PSO 2	PSO 3	PSO4	PS O5	PSO 6	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO 1	3	1	0	1	0	0	3	2	0	0	2	0	1
CO 2	3	1	0	0	0	0	3	2	0	0	0	0	2
CO 3	3	0	3	0	0	0	3	2	0	0	0	0	1
CO 4	3	1	0	0	0	0	3	2	0	0	0	0	1
CO 5	3	0	1	0	0	0	3	2	0	0	0	0	1
CO 6	3	0	0	0	1	1	3	2	0	0	0	0	1
CO 7	3	0	0	0	0	1	3	2	0	0	0	0	2

Correlation Levels:

Level	Correlation
0	Nil
1	Slightly / Low
2	Moderate / Medium
3	Substantial / High

Assessment Rubrics:

- Quiz / Discussion / Seminar
- Internal Theory/Practical Exam
- Assignments /Viva
- End Semester Exam (70%)

Mapping of COs to Assessment Rubrics

	Internal Theory/ Practical Exam	Assignment /Viva	Practical Skill Evaluation	End Semester Examinations
CO 1	✓	✓		✓
CO 2	✓	✓		✓
CO 3	✓	✓		✓
CO 4	✓	✓		✓
CO 5	✓	✓		✓
CO 6	✓	✓		✓
CO7		✓	✓	